

Supplementary Material for Topology-Aware Differentiable Triangle-Soup Reconstruction via Persistent Homology

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A. The allocation-channel study, in brief

This appendix reports, with their numbers, the three findings of our allocation-channel study — controlled resampling-prior experiments on the same pipeline, scenes, and budgets, conducted before moving topology into the objective — that this paper builds on.

(1) Geometric metrics are topology-blind. Three constructed cases each pit a topologically correct candidate A against a wrong one B whose geometry is bisection-matched, making Chamfer equal by construction (40k samples, metrics normalized by the ground-truth bounding diagonal; Table S1). The discriminating-dimension bottleneck separates every pair — by 35–40× in the H0/H2 cases — while in the first two cases Hausdorff₉₅ actually prefers the topologically wrong candidate. Chamfer cannot tell a benign bump from a destroyed void; the persistence metric can.

(2) The prior arms, and why voids are “width-primary”. The study biases DiffSoup’s own resampler with a pre-computed importance field built from the target’s significant features. Each significant feature f (lifetime p_f above the significance threshold) contributes Gaussian kernels at the coordinates C_f of its birth/death-simplex vertices — the loop-filling triangle, void-filling tetrahedron, or component-merging edge — sized by the feature’s own death scale:

$$\phi_d(q) = \sum_{f: \dim f=d} \sum_{c \in C_f} w_f e^{-\|q-c\|^2/(2\sigma_f^2)}, \quad w_f = p_f/s^2, \\ \sigma_f = s \operatorname{clip}(\sqrt{\operatorname{death}_f}, 3r_{\text{med}}, \tfrac{1}{2}\operatorname{diag}), \quad (1)$$

evaluated on a dense target-surface sample, normalized per dimension to $[0, 1]$ (99.5th percentile), combined by max; a query takes its nearest surface sample’s value.

The spread knob s widens each kernel while preserving its injected surface mass ($\sigma \rightarrow s\sigma$, $w \rightarrow w/s^2$, so $w\sigma^2$ is s -invariant): $s=1$ is the concentrated prior (B2), $s=3$ the spread prior (B4) — the field C5 reuses. The field enters resampling at strict budget parity through two levers: prune protection — faces drop in order of $\text{vis} + \lambda Q_{0.9}(\text{vis}) \cdot \text{imp}$, removing exactly the required count — and respawn bias — top-importance visible faces subdivide first, then the protective order re-prunes to the cap. Remaining arms: B0 baseline; B1 = B2 applied at initialization only; B3/B5 random-center non-topological controls width-matched to B2/B4 respectively. On the void class (Table S2; $N=1200$, five paired seeds) B4 cuts the bottleneck 33–53% below baseline — but the width-matched random control B5 recovers most of that gain, leaving only a small, shape-dependent topological residual (cylinder 4.6σ , sphere 2.2σ , cube tie); one-shot placement washes out entirely ($B1 \approx B0$, .0546 vs .0535 on the sphere). Hence the verdict quoted in the main paper’s introduction: the prior’s topological value is carried largely by its spatial width.

(3) No prior shape repairs loops. On the torus ($N=700$, five paired seeds, true $\beta_1=2$) every arm fails the topology-specificity test: the best condition is the non-topological wide control B5 (.0267), ahead of spread-topological B4 (.0316), baseline (.0409), and narrow random B3 (.0397); the concentrated topological field B2 is worst (.0425) and manufactures phantom handles — final significant-H1 count 4.4 against the true 2.0, with the worst Chamfer (1.32 vs baseline 1.01). Spreading merely stops the harm; it never beats the random control. That failure is the opening premise of the main paper, and the class the loss channel wins there (main paper §5.1, Table 1).

Table S1. Blindness probes: A topologically correct, B wrong, geometry bisection-matched so Chamfer is equal by construction. Bottleneck to ground truth in the discriminating dimension.

case (disc. dim)	Cham.% A = B	bott. A	bott. B	ratio
thin handle, genus 0→1 (H1)	0.916	≈0	.0302	—
bridge merges two components (H0)	0.628	.0014	.0574	40×
punctured enclosed void (H2)	0.977	.0040	.1385	35×

Table S2. Allocation study, void class (H2), $N=1200$, five paired seeds: tail bottleneck to target (lower is better). Spread beats concentrated; the width-matched random control (B5) recovers most of the spread gain.

	B0	B2 conc.	B3 narrow rand.	B4 spread	B5 wide rand.
sphere	.0535	.0438	.0440	.0357	.0374
cube	.0579	.0409	.0403	.0273	.0270
cylinder	.0545	.0440	.0477	.0360	.0402

Table S3. Significant bars (H1 / H2) on the clean target surfaces vs. sampling density M ; floor = $6r_{\text{med}}(M)$, seed 0.

	$M=512$	$M=1024$	$M=2048$	$M=4096$
sphere	0 / 1	0 / 1	0 / 1	0 / 1
torus	1 / 0	1 / 0	2 / 0	2 / 1
double torus	0 / 0	2 / 0	2 / 0	4 / 0

B. Diagnostics: density sensitivity and diagram trajectories

Sampling density vs. measurability. Table S3 counts the significant bars of the clean target surfaces across sampling densities under the self-calibrating floor $6r_{\text{med}}(M)$ of the main paper’s §6. It confirms the three density claims made in the paper: the sphere’s void clears the floor at every tested density; the torus needs $M=2048$ before its second loop is measurable, and its own void becomes measurable only at $M=4096$ (hence the H1 restriction at $M=2048$); and the double torus’s tube loops cross the floor between $M=2048$ and $M=4096$ — consistent with the $M \gtrsim 3 \times 10^3$ estimate there — so all four loops are measurable at $M=4096$.

Diagram trajectories: the pinning mechanism, visualized. Fig. S1 shows the live H2 diagrams of one sphere seed under C0/C1/C2 at four training steps. All three arms start with the void essentially on target (initialization samples the SfM point cloud), and photometric training alone drags it away (the baseline settles ≈.06 below). The loss arm dips identically while the ramp is inactive, then pulls the void back to within ≈.016 of

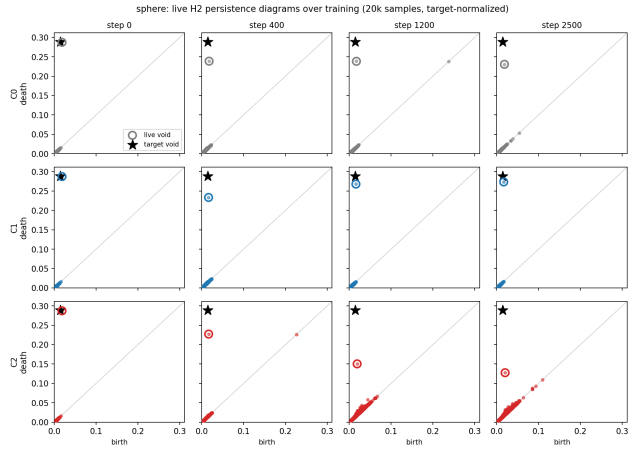


Figure S1. Live H2 persistence diagrams over training (one sphere seed; 20k samples, target-normalized; star = target void bar, circle = most persistent live bar). Top, baseline: the void drifts off target. Middle, loss: the void returns to the target once the ramp activates. Bottom, repulsion control: the void is pinned and dragged away while near-diagonal phantom bars accumulate.

target — matching its measured tail. The control arm is qualitatively different: the void’s death is dragged progressively down (≈.23 → .15 → .13) while a carpet of phantom mid-persistence bars accumulates near the diagonal — the repulsion-equilibrium signature discussed in the main paper’s §6.

The genus-2 stretch case, in full. On the double torus every arm — baseline, loss, control — returns a bottleneck of exactly .0264 (sd 0 over five seeds; raised from two in revision, identical values throughout), and Wasserstein is equally pinned (.0527–.0528): the two tube loops are unexpressable at feasible budgets (replicating the allocation study’s floor, §A) and dominate both diagram metrics regardless of guidance. The loss adds no phantoms ($\# \text{sig } \beta_1 = 2$ every run) at marginally best Chamfer.

Sensor-noise internals (C7/C7h). Recruitment absorbs the injected noise: at $\sigma=1\%$ the torus’s second loop is sub-threshold at every refresh, with zero un-

Table S4. Tom-yum pot challenge inventory: what makes the shape hard, and the measured consequence in this study.

challenge	measured consequence
raw artist export (6,318 vertices): 9 open shells, mixed quad/12-gon faces, 50 boundary edges after exact weld, 232 non-manifold junction edges	no trustworthy ground-truth topology as exported; ingest-and-certify required (weld, offset-solidify into a thin shell via an exact unsigned distance field + marching cubes at iso ϵ , drop 804 enclosed pocket shells)
certified shell: one body, genus 3, $\beta=(1,6,1)$ exact, 190,990 vertices	ground truth known by certificate (exact simplicial homology; watertightness cross-check; bit-identical rebuilds), not by construction
six thin handle/vent loops	every H1 loop sits below the significance floor at every working density — H1 is unusable as the loss observable
narrow flue mouth, capping at small α	the flue interior reads as an enclosed on-axis chamber — an H2 void, first measurable at $M=8192$ (margin 1.20–1.23 $\times$, five sampling seeds); the floor rule picks class and density prospectively (Table S5)
in-loop outcome	C1 cuts the void’s error 3.9 $\times$ at Chamfer parity; the repulsion control destroys it ($\beta_2 1 \rightarrow 0$, all seeds, 1.36 $\times$ worse Chamfer) — main paper Table 2

reached targets. Calibration is the noisiest stage (λ_{peak} spreads $\approx 3\times$ across seeds), yet tails stay tight — consistent with the flat ρ response reported in the main paper.

The tom-yum pot: challenge inventory and prospective floor protocol. The generality wave’s artist mesh (main paper §4, §5.2) compounds the failure modes real scans will bring. Table S4 inventories them with the measured consequence of each — every row is backed by the ingest certificate or by a measurement in this study, none is asserted. Table S5 records the staircase behind the flue row: the certified shell’s significant-bar signature across sampling density, which fixed the observable class (H2) and the bundle density ($M=8192$) before any training run.

C. Additional result figures and the repair gate

Fig. S2 shows the generality-wave training trajectories (main paper §5.2, Table 2); Fig. S3 the per-seed tail spread behind the sphere row of main-paper Table 1; Fig. S4 the torus significant-count check behind §5.1’s zero-phantom-handles claim.

Table S5. Tom-yum pot staircase (seed 0; the $M=8192$ margin spans seeds 0–4): significant-bar signature ($\beta_0, \beta_1, \beta_2$) of the certified shell vs. sampling density M , floor = $6r_{\text{med}}(M)$. The bundle is built at the first density where the discriminating feature clears ($M=8192$: exactly one significant bar, H2, stable to re-sampling at $\sim 10^{-5}$).

	$M=2048$	$M=4096$	$M=8192$	$M=20000$
sig. bars	(1, 0, 0)	(1, 0, 0)	(1, 0, 1)	(1, 0, 1)

The stage-3a gate (main paper §5). Fig. S5 shows three of the four repair probes run before integration: the loss alone (no photometric term, Adam directly on point positions) must repair defective clouds. The predicted location-blind pathology of recruitment did not materialize, for a mechanism reason worth stating: tearing a 2-manifold shell cannot grow a component bar past the local detour scale (the tug is abandoned), while cutting the 1-D bridge raises the recruited bar’s death monotonically toward target — the runaway concentrates on the only profitable direction.

D. A decade of reconstruction, cut by optimization philosophy

The main paper’s related work argues one sentence that this appendix substantiates in breadth: reconstruction methods of the differentiable decade, whatever their representation, optimize geometric objectives (photometric fidelity plus geometric regularizers) or steer sampling under such an objective — topology, when correct, is correct by construction, by extraction, or by accident, never because the training objective measured it. Table S6 tabulates representative methods under this cut, including the nearest concurrent topology-aware entries; the subsections survey each family and the adjacent literatures this paper draws machinery or discipline from.

D.1. Geometry-oriented: implicit fields

Occupancy and signed-distance learning deliver closed surfaces by construction [54, 61, 62], and a fitting stack matured around the level set: Eikonal regularization [25], surface rendering [60, 83], SDF-based volume rendering [79, 84], geometric priors [87], high-resolution and accelerated variants [51, 57, 80], meshing for deployment [85], and kernel-based point-cloud fitting [39]. Across all of them the training objective is photometric or geometric; the guarantee of consistent topology comes from the representation, its correctness (the right genus for the scene) is emergent, and the price is dense field evaluation plus post-hoc extraction — the

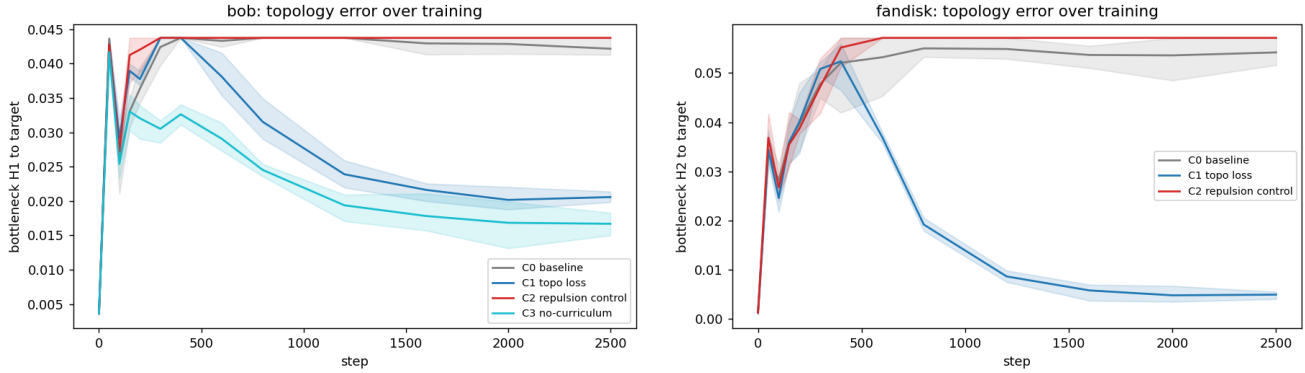


Figure S2. Generalization trajectories (line = seed mean, band = seed min-max). Left: bob (genus 1) — the H1 result off the analytic family: the loss halves the loop error with #sig H1 = 2 in every seed, while the control rides the baseline after collapsing a loop. Right: fandisk (CAD) — the largest effect measured in this study (10.4 $\times$).

Table S6. Representative reconstruction methods by optimization philosophy. “Topology in objective?” asks whether the training objective measures a topological quantity of the reconstructed surface (not whether the representation constrains topology, and not image-space summaries). Bottom block: the nearest topology-aware concurrent work (§D.7).

method	repr.	philosophy	topology in objective?	topology from
NeRF [55]	density field	geometry	no	emergent
OccNet [54]	occupancy	geometry	no	construction (closed iso)
DeepSDF [61]	SDF	geometry	no	construction (closed iso)
NeuS [79]	SDF	geometry	no	construction (closed iso)
Neuralangelo [51]	SDF	geometry	no	construction (closed iso)
Pixel2Mesh [78]	template mesh	geometry	no	frozen at template genus
Point2Mesh [30]	template mesh	geometry	no	frozen at template genus
Mesh R-CNN [24]	voxels $\rightarrow$ mesh	geometry	no	fixed by voxelization
DMTet [71]	grid + marching	geometry	no	extraction over the grid
nvdiffr [58]	grid + marching	geometry	no	extraction over the grid
FlexiCubes [72]	grid + marching	geometry	no	extraction over the grid
Shape-as-Points [63]	points $\rightarrow$ implicit	geometry	no	construction (closed iso)
AtlasNet [26]	patch soup	geometry	no	unmeasured
DSS [86]	point soup	geometry	no	unmeasured
3DGS [47]	Gaussian soup	geometry	no	unmeasured
2DGS [37]	surfel soup	geometry	no	unmeasured
SuGaR [27]	Gaussians $\rightarrow$ mesh	geometry	no	post-hoc extraction
MiLo [28]	Gaussians+mesh	geometry	no	in-loop extraction, unmeasured
Triangle Splatting [31]	triangle soup	geometry	no	unmeasured
Radiant Tri. Soup [8]	triangle soup	geometry	no	unmeasured (proximity forces)
Mini-Splatting [19]	Gaussian soup	sampling	no	unmeasured (budgeted resample)
DiffSoup [76]	triangle soup	geometry	no	unmeasured
DiffSoup + prior (§A)	triangle soup	sampling	no (steers only)	unmeasured; fails on loops
Topology-GS [73]	Gaussian soup	topology (image)	rendered images only	surface unmeasured
Gao et al. [22, 23]	watertight mesh	geometry	no (topology a priori)	template genus / camera prior
STITCH [41]	implicit (points)	topology	yes (β_0 , live surface)	construction + constraint
this work	triangle soup	topology	yes (live-surface persistence)	measured, optimized, budgeted

cost regime the soup family exists to avoid.

D.2. Geometry-oriented: extraction, grids, and templates

Where the surface is explicit, its topology is decided by an extraction algorithm or frozen by a template. Marching Cubes and Dual Contouring fix per-cell con-

nectivity by convention [43, 52]; Poisson reconstruction extracts an indicator iso-surface, watertight but with emergent genus [44, 45]; differentiable iso-surfacing makes extraction trainable without making topology a target [21, 58, 65, 71, 72]; learned tessellations move the per-cell decision into a network [14, 15]; and template pipelines fix topology up front and then only

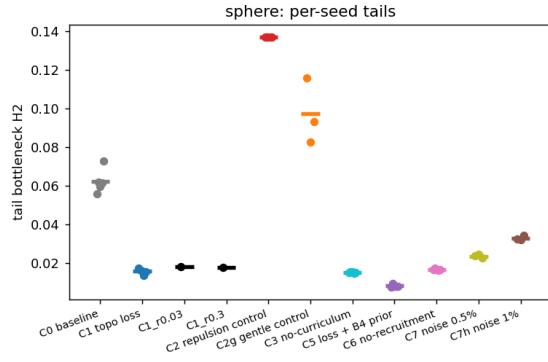


Figure S3. Sphere, per-seed tail bottlenecks (dots) with arm means (bars). Every loss arm sits far below baseline with tight seed spread; the two control strengths bracket baseline from above; the ρ ablation arms are indistinguishable from $\rho=0.1$.

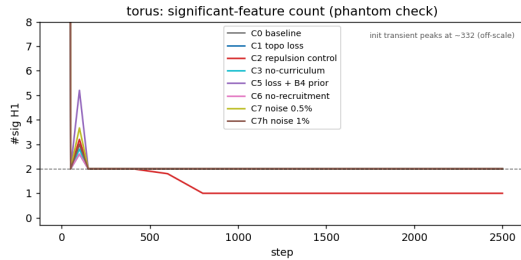


Figure S4. Torus phantom check (main paper §5.1): significant-H1 count over training (seed means; dashed line: true $\beta_1=2$). After the brief resampling transient near step 100, every loss arm sits exactly on two significant loops — zero phantom handles — while the repulsion control (C2) collapses a loop by step ~ 800 and never recovers it.

deform [24, 30, 78]. Robust meshing gives the same guarantee-by-construction for simulation inputs [36]. In every case topology is an input or an artifact of the algorithm — measured nowhere, optimized never.

D.3. Geometry-oriented: point splatting and Gaussian soups

The connectivity-free line runs from EWA surface splatting [92] through neural point rendering [2, 67, 82] to the Gaussian-splatting era [47, 88] — Pulsar’s abstract advertises the family’s freedom from mesh “topology problems” explicitly [49]. Its surface-minded members regularize geometry ever harder — flattened surfels [18, 37], planar compression [13], opacity-field extraction [89], alignment-driven meshing [27] — and MIlO now extracts a mesh differentially at every training iteration [28]; triangle soups follow the same pattern [8, 31, 76]. Yet even with connectivity in the loop,

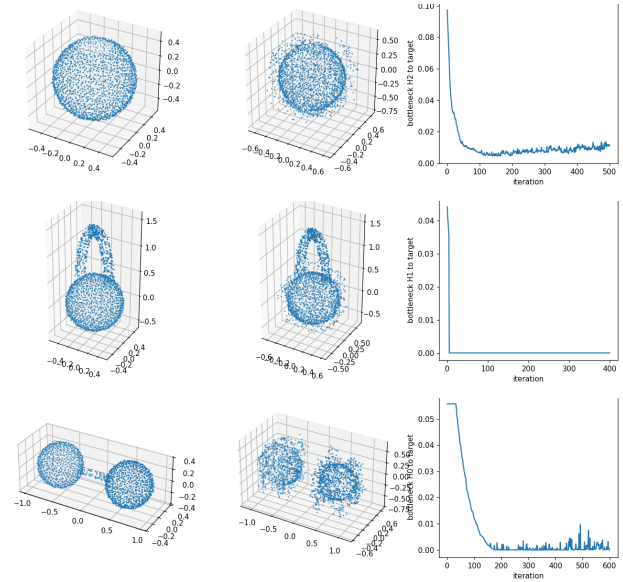


Figure S5. The stage-3a gate: the loss alone (Adam on raw points, no photometric term) repairs defective clouds; each row shows initial cloud, final cloud, and bottleneck-to-target trajectory. Top (T1): punctured sphere — the matched birth term closes the void ($8.8\times$). Middle (T2): spurious handle — the diagonal term kills the extra H1 bar. Bottom (T4): bridged merge with no significant live H0 bar — recruitment severs the bridge ($450\times$, $\beta_0 1\rightarrow 2$), the case where plain optimal matching provably has zero gradient. (T3, mis-spaced components, $1229\times$: omitted for space.)

no member of the family measures a topological quantity of what it builds: the objectives remain photometric fidelity plus geometric regularity.

D.4. Sampling-oriented: steering where a fixed objective spends

The second occupied family changes where primitives go, not what the objective asks of them: split-and-prune densification in 3DGS [47], budget-constrained Gaussian resampling [19], importance-driven respawn in DiffSoup [76] — and the topology-inspired allocation priors of our own study (§A), which put the family’s implicit hypothesis to a controlled test and returned the null result the main paper builds on. Steering allocation is a real lever for geometric coverage; the study shows it is structurally the wrong lever for one-dimensional topology.

D.5. Topology as machinery: persistence in and around learning

The ingredients for topology-aware objectives have been maturing for two decades: persistence computation at scale [5, 53, 91], practical diagram met-

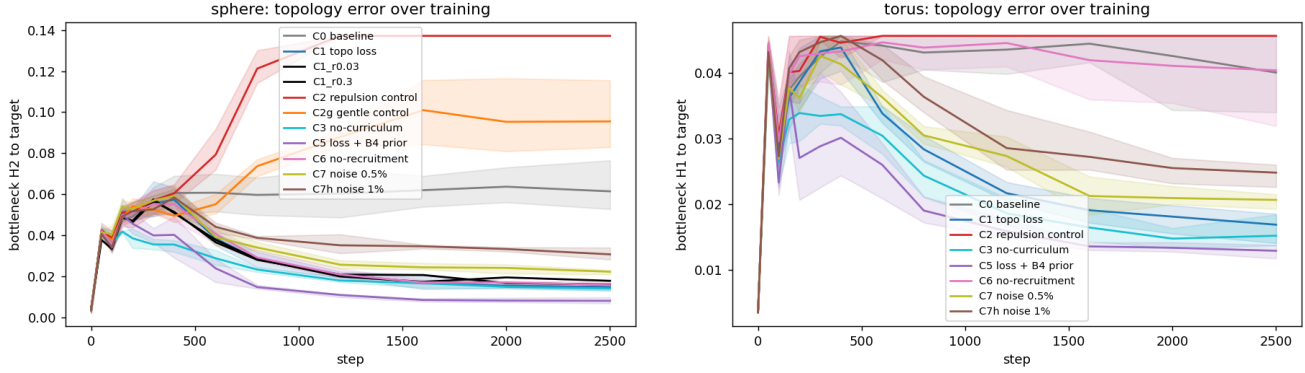


Figure S6. Topology error over training: bottleneck to target in the discriminating dimension (line = seed mean, band = seed min-max). Left, sphere (H2): every loss arm (C1, C3, C5, C6, both ρ variants, noise pair C7/C7h) converges to a fraction of baseline; the norm-matched repulsion control C2 is destructive, and its gentle variant C2g still ends worse than baseline. Right, torus (H1), the class no prior won: same ordering, C5 best; noise arms descend shallower but never approach baseline; the no-recruitment ablation C6 rides it.



Figure S7. The tom-yum pot: an artist-authored Blender model (raw export: 9 open shells, 232 non-manifold junction edges) ingested and certified to a single genus-3 shell, $\beta=(1,6,1)$ exact (main paper §4). Right: cutaway — the narrow flue mouth caps at small α , so the flue interior reads as an enclosed chamber: the on-axis H2 void the floor rule selects at $M=8192$ (all H1 loops sit below the floor).

rics [46], stability theory [17], and vectorized representations [1, 7] on the computational side; trainable diagram layers [9, 32], filtration learning [33], and general differentiability frameworks [10, 50, 59] on the gradient side (surveyed in [11, 12]). In-loop topological losses already win in adjacent domains — representation learning [56], tubular and topologically faithful segmentation [16, 35, 74, 75], microscopy volume prediction [77], point-cloud repair [6, 64] — and differentiable non-persistence signals exist too [66]. What had not happened is the transfer of this machinery into image-based surface reconstruction.

D.6. Topology outside the loop: repair and benchmarks

Before differentiable pipelines, topology was corrected after the fact: spurious-handle removal [29, 81], general mesh repair [3, 42, 70] (surveyed in [4]), and soup-

to-manifold conversion [36, 38]; Thingi10K quantifies how pervasive soup-like defects are in the wild [90] — our tom-yum pot’s raw export is a typical specimen. The evaluation culture matches: standard pipelines and benchmarks — COLMAP [68, 69], PMVS [20], DTU [40], Tanks and Temples [48] — score accuracy and completeness by point distances; none carries a topological-correctness metric, the blindness our Table S1 quantifies on controlled probes.

D.7. The empty cell, and its nearest neighbours

Concurrent work has begun to press on the gap from several directions, and the distinctions matter. Topology-GS [73] puts a persistence term inside Gaussian-splatting training — but on barcodes of rendered images, a topological-perceptual loss; the surface’s diagram is never measured, and densification adds primitives rather than budgeting them. The high-genus inverse-rendering line [22, 23] reconstructs meshes whose genus is guaranteed by a Gauss–Bonnet-matched template or guides camera placement with homology computed once on a reference shape — topology as a prior, assumed rather than measured, with ground-truth genus required up front. STITCH [41] does measure a differentiable persistence quantity on a live implicit surface — enforcing a single connected component — but from point-cloud input: no images, no primitives, no budget. Topology also enters generation as conditioning features [34], again outside the reconstruction objective. Partition the three requirements — (i) a topological quantity of the evolving surface measured differentiably in the objective, (ii) image-based reconstruction through a renderer, (iii) a fixed primitive budget — and each neighbour holds at

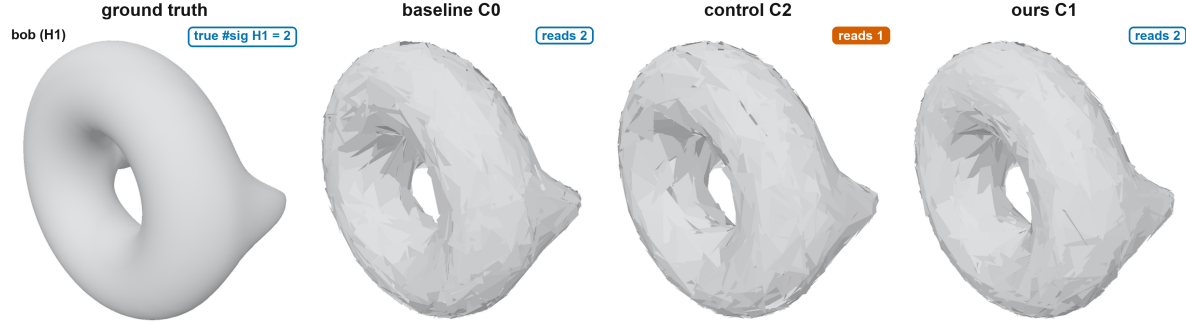


Figure S8. The loop-class row of the main paper’s Fig. 1 matrix: bob (genus 1, H1). The repulsion control’s reconstruction reads one significant loop against the true two (β_1 2→1, all seeds) at 2× worse Chamfer, while baseline and loss both read two — the loss at half the baseline’s tail error (main Table 2).

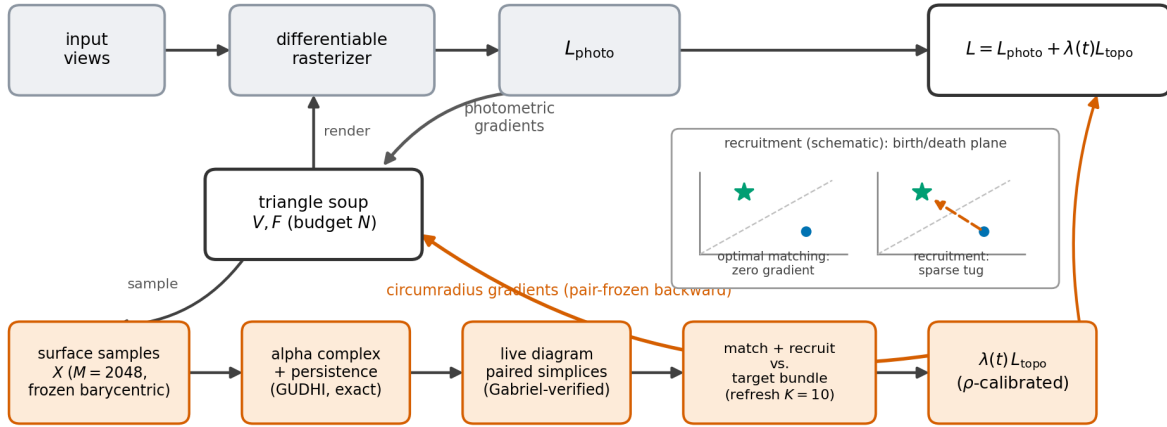


Figure S9. The system, schematically. DiffSoup’s photometric path (gray) is untouched; the topological branch (orange) samples the live soup ($M=2048$, frozen barycentric weights), computes exact alpha-complex persistence, matches and recruits against the fixed target bundle (refresh every $K=10$ steps and after every resample), and back-propagates through the pair-frozen circumradius re-expressions into the same vertices (main §3). Inset (schematic): an absent target bar (star) receives no gradient under optimal matching (main Lemma 1); recruitment attaches the nearest unclaimed live bar — the sparse tug.

most one; none holds all three. That cell — the bottom row of Table S6 — is what the main paper fills.

E. Per-seed values and calibration sweeps

The main paper’s reporting policy (§4) quotes Welch σ separations only where both arms ran five seeds; every three-seed verdict instead rests on disjoint per-seed ranges. Table S7 lists the underlying per-seed tail bottlenecks for the generality wave and the three-seed analytic shapes. In every row except rocker-arm’s (the no-headroom null, main §5.2) the largest C1 seed is well below the smallest C0 seed; identical values across seeds — C2’s, and the group wave’s pinned C0s — are genuine (the unmatched-target bound, main paper

§§5.2, 6; suppl. §G), not rounding.

Table S8 tabulates the two calibration-knob sweeps whose verdicts the main paper states in §3.4 and §5.1: the ρ response is flat over a 10× range, and the ramp window is immaterial — all alternatives land within seed noise of the default.

F. The open-surface probe

Two open bowls are cut from the analytic sphere’s ground-truth mesh by removing a polar cap (measured openings 0.14 R and 0.50 R), then certified by measurement as disk topology: one shell, exactly one boundary loop, edge-manifold, consistently oriented, Euler characteristic 1. The pre-registered bundle rule (the pot’s

Table S7. Per-seed tail bottlenecks (seeds 0/1/2, run order) behind the three-seed verdicts. Spot’s C1 value repeats across seeds at this precision; the pot rows are the quicklook values behind main Table 2.

shape (dim)	C0	C1 loss	C2 control
spot (H2)	.0429 / .0584 / .0550	.0237 / .0237 / .0237	.0652 ×3
bob (H1)	.0429 / .0413 / .0437	.0200 / .0207 / .0216	.0437 ×3
fandisk (H2)	.0540 / .0515 / .0559	.0058 / .0055 / .0042	.0571 ×3
tom-yum pot (H2)	.0218 / .0223 / .0223	.0059 / .0056 / .0056	.0223 ×3
rocker-arm (H1)	.0088 / .0082 / .0079	.0090 / .0070 / .0100	.0112 / .0112 / .0115
eight (H1)	.0249 ×3	.0169 / .0175 / .0142	.0249 ×3
armadillo (H2)	.0511 ×3	.0124 / .0125 / .0123	.0511 ×3
horse (H2)	.0408 ×3	.0124 / .0083 / .0118	.0408 ×3
cube (H2)	.0560 / .0551 / .0636	.0071 / .0064 / .0086	.1346 ×3
two spheres (H0)	.0060 / .0074 / .0062	.0007 / .0009 / .0005	.0009 / .0009 / .0006

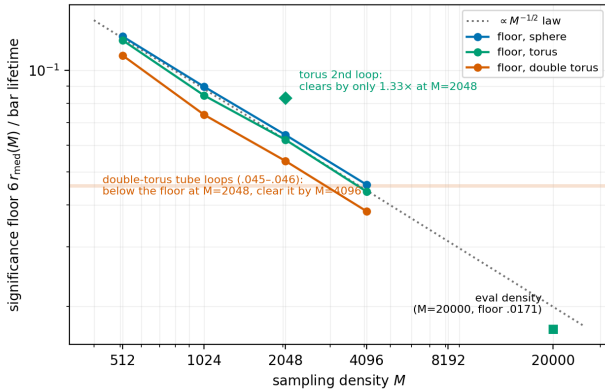


Figure S10. The measurement floor, visualized (all values from the recorded density sweep behind Table S3 and main §6). The floor $6 r_{\text{med}}(M)$ falls as $M^{-1/2}$ (dotted: the law anchored at the torus $M=2048$ point, reaching the $M=20000$ evaluation floor .0171), while a feature’s lifetime is fixed — a feature becomes measurable where the falling floor crosses it. The double torus’s tube loops (.045–.046 band) sit below the floor at $M=2048$ and clear it by $M=4096$, matching Table S3’s counts; the torus’s second loop clears by only $1.33\times$ at $M=2048$ (lifetime marked at $1.33\times$ the .0623 floor) — the thin margin that live-cloud noise consumes (main §6, C6).

Table S8. Calibration-knob sweeps (sphere, tail bottleneck). Top: ρ response, matched seed-0 sweep — the five-seed $\rho=0.1$ arm is main Table 1’s C1 (.0156±.0013). Bottom: ramp-window pilot, three seeds each.

ρ (seed 0)	0.03	0.1	0.3
tail	.0181	.0172	.0177
ramp window	10–40%	20–50% (default)	30–60%
tail	.0154±.0017	.0156±.0013	.0179±.0019

Table S9. Probe staircase (seed 0): top bar lifetime over the floor, per dimension. Both bowls read exactly (0,0,1) significant bars at every density — one H2 void, no significant H1 — so the table gives the margins.

top bar / floor	$M=2048$	$M=4096$	$M=8192$
narrow: H2 / H1	3.79× / 0.47×	5.50× / 0.57×	7.82× / 0.45×
wide: H2 / H1	2.34× / 0.42×	3.50× / 0.51×	4.94× / 0.46×

expectation. We predicted the wide bowl would lose its enclosed void to the rim; instead both bowls read exactly one significant H2 bar at every tested density (the mouth caps at small α , so the interior reads as a chamber — the pot’s mechanism on an open surface), while the designed rim’s H1 bar never clears the floor (Table S9). The rule therefore sets H2 at $M=2048$ for both, budget $N=1200$, loss restricted to H2 (the torus precedent for sub-floor features). A designed rim is a real feature of the open target, distinct from the noise-born boundary bars real scans will add — the bar-filtering step named in main §7 concerns the latter.

The C-matrix (C0/C1/C2, three seeds, protocol otherwise unchanged) passes the verdict rule on both bowls at better-than-baseline Chamfer (Table S10): the loss cuts the void’s error $4.2\times$ (narrow) and $4.1\times$ (wide) with disjoint per-seed ranges, inside the study’s $1.5\text{--}10.4\times$ span. The control replicates its closed-surface failure modes: on the narrow bowl it pins the void at its repulsion equilibrium ($2.1\times$ worse than baseline, identical across seeds) at $3.5\times$ worse Chamfer; on the wide bowl it erases the void in two of three seeds ($\beta_2 1\rightarrow 0$) at $3.1\times$ worse Chamfer. Every C0/C1 seed reads the correct count.

applies unchanged: the density staircase decides the observable and density — and it overruled our own

Table S10. Open-surface probe C-matrix: tail bottleneck (mean $\pm$ sd, 3 seeds; per-seed values in run order beneath); $\times$ = reduction vs C0. Both shapes pass the verdict rule; no Welch σ at $n=3$ (main §4).

shape	C0	C1 loss	C2 control
bowl _{narrow}	.0588 $\pm$.0035	.0138 $\pm$.0013 (4.2 $\times$)	.1249 (pinned)
per-seed	.0619/.0550/.0593	.0137/.0126/.0151	.1249 $\times$ 3
bowl _{wide}	.0560 $\pm$.0098	.0137 $\pm$.0003 (4.1 $\times$)	.0770 (β_2 1 $\rightarrow$ 0, 2/3)
per-seed	.0448/.0602/.0630	.0141/.0136/.0135	.0770 $\times$ 3

G. The group wave: expanding the external set to two groups

The generality wave of main §5.2 grew from four external meshes to eight, organized as the study’s two observable classes — the loop group (H1: bob, rocker-arm, eight) and the void group (H2: spot, fandisk, the tom-yum pot, armadillo, horse) — under a pre-registered plan (GROUP_WAVE_PLAN.md, shipped with the release bundle) that fixed the candidate pool, the selection rule, the run protocol, and the group statistic before any training run. Per-seed tails for the new shapes join suppl. Table S7; this section records where the shapes came from and how they were selected.

Pool, certificates, honest exclusions (Table S12). Candidates were drawn from a standard test-model collection (alecjacobson/common-3d-test-models, pinned commit) plus the CGAL repository’s genus-2 eight; the release bundle ships fetch scripts with pinned commits and the certificates, not the meshes — redistribution terms remain the original repositories’. Every candidate passed or failed the pot’s mesh certificate (main §4: exact simplicial homology cross-checked against per-component Euler characteristics, watertightness, orientation) as-is — failures were excluded, never repaired, keeping the wave’s class “watertight benchmark” (the pot remains the study’s one ingest-and-solidify story). Five of eleven failed: open boundaries or non-manifold junctions in beast, cow, max-planck, ogre, and the Utah teapot.

Staircase selection, before training (Table S11). The density staircase (the pot’s pre-registered rule, unchanged) fixed each certified shape’s observable class and working density: rocker-arm (genus 1) enters the loop group on the torus precedent — one of its two loops clears the $M=2048$ floor (0.0929 vs 0.0613), the second (0.0204) reaches only 1.15 $\times$ the floor even at $M=20,000$, and the void is sub-floor at 2048, so the loss is H1-restricted; eight (genus 2) enters on the pot’s precedent — the full exact (1, 4, 1) is significant from

Table S11. Group-wave staircases (seed 0): significant Betti counts vs sampling density for the four selected shapes; bold = the pre-registered working density. rocker-arm’s second loop and 2048-density void never clear the floor (loss H1-restricted, torus precedent); eight’s full (1, 4, 1) holds from 4096 (bundle at 8192, pot precedent).

sig ($\beta_0, \beta_1, \beta_2$)	$M=2048$	4096	8192	20000
rocker-arm	(1,1,0)	(1,1,1)	(1,1,2)	(1,2,2)
eight	(1,2,1)	(1,4,1)	(1,4,1)	(1,4,1)
armadillo	(1,0,1)	(1,0,1)	(1,0,1)	(1,0,1)
horse	(1,0,1)	(1,0,1)	(1,0,1)	(1,0,1)

Table S12. Candidate disposition (selection fixed before training; plan file in the release bundle). Certificate failures are excluded, never repaired.

candidate	origin	certificate	decision
rocker-arm	INRIA / AIM@SHAPE	pass, genus 1	loop group
eight	CGAL data (CC0-class)	pass, genus 2	loop group
armadillo	Stanford scan rep.	pass, genus 0	void group
horse	CyberWare / GaTech	pass, genus 0	void group
cheburashka	I. Baran (Pinocchio)	pass, genus 0	excluded: 2nd capping H
eight (dense)	Directional (lib.)	pass, genus 2	excluded: license; corrob
beast	Autodesk	fail: non-manifold	—
cow	Viewpoint/Sun	fail: odd χ	—
max-planck	MPI	fail: 161 open edges	—
ogre	K. Ritchie	fail: 200 open edges	—
teapot	M. Newell	fail: 160 open edges	—

$M=4096$, and the bundle is built at $M=8192$ where the fourth loop clears 1.69 $\times$ and the void 2.6 $\times$; armadillo and horse (genus 0) read the exact (1, 0, 1) at every tested density (voids 1.68 $\times$ /1.42 $\times$ over the 2048 floor). Two certified candidates were set aside at this stage, before any training: cheburashka qualifies at 2048 but grows a second significant capping H2 bar from $M=8192$ (0.0300 vs floor 0.0271; 0.0312 vs 0.0171 at 20k) — a pot-flue-type concavity reading that would make the evaluation-density phantom check ambiguous — and a denser retessellation of eight (license unstated at its source) whose staircase independently corroborates the (1, 4, 1) observable at $M=8192$.

Outcome identities behind the pins. The three seed-exact baseline values of main Table 2 are unmatched-target bounds, not coincidences: eight’s C0/C2 pin 0.0249 $\approx$ half the third loop’s ground-truth lifetime at the evaluation density (0.0498/2), armadillo’s 0.0511 = 0.1022/2, horse’s 0.0408 $\approx$ 0.0817/2 — the bottleneck cost of leaving that target bar unmatched (main §6’s pin mechanism, previously seen in the pot’s control arm).

H. Robustness of the reporting choices

The floor multiplier is not load-bearing. The significance floor $k r_{\text{med}}(M)$ uses $k=6$ throughout. Recom-

Table S13. Floor-multiplier windows per shape (recorded GT clouds at the protocol bundle density): the tightest included margin and, where the protocol excludes sub-floor features, the largest excluded margin — each decision holds while k stays below/above $6\times$ that margin. Global window: $k \in (5.12, 6.77)$.

shape	tightest IN ($\times$ floor)	largest OUT ($\times$ floor)
sphere / cube	3.86 / 2.71	—
torus	1.33 (loop 2)	0.83 (void)
two spheres	2.61	—
double torus	2.23	0.85 (tube loops)
spot / fandisk	1.87 / 1.50	—
bob	1.13 (void)	—
tom-yum pot	1.22	0.46 (loops)
rocker-arm	1.52	0.74 (void)
eight	1.69 (loop 4)	—
armadillo / horse	1.68 / 1.42	—
bowls (narrow/wide)	3.79 / 2.34	0.47 / 0.42 (rims)

— an $11.0\times$ cost for $9.8\times$ the points, consistent with the near-linear $O(M \log M)$ scaling. GPU memory: a single-seed probe (torus, 2,500 steps, whole-GPU used memory sampled at 400 ms) peaks at 3,649 MiB for the baseline and 4,035 MiB with the loss (+386 MiB); the persistence computation itself runs on the CPU (GUDHI), off the GPU budget.

putting every documented bundle decision from the recorded ground-truth clouds (protocol densities, bundle seed) gives each feature’s flip point — the k at which an included feature would leave the bundle or an excluded one would enter — and every decision in the study is unchanged for $k \in (5.12, 6.77)$ (Table S13). The two binding features are exactly the ones the paper already flags as marginal: the double torus’s sub-floor tube loops ($0.85\times$ the $k=6$ floor) bound it below, bob’s fat-tube void ($1.13\times$) bounds it above. The verdict metric itself (bottleneck to the target diagram) contains no threshold; k only gates which bars enter bundles and counts.

The pair-frozen plan is stable across refreshes (Fig. S11). Every run since the refresh-log schema records the plan’s composition (matched / diagonal / recruited / target-unreached, per dimension) at each of its ~ 200 refreshes. Three regimes cover the study: the torus without recruitment (C6) shows the chronic gap the ablation collapses on — one matched loop and one unreached target at every refresh; on the genus-2 eight, recruitment claims the two sub-threshold loops from the first active refresh and hands them over to the matched set as training makes them significant (matched $2 \rightarrow 3-4$ late, recruited $2 \rightarrow 1-0$, unreached 0 throughout); the tom-yum pot settles to a stable one-bar matched plan within the first ~ 250 active steps. No plan oscillation beyond single-bar hand-overs is observed, and the per-refresh Gabriel-failure count is zero in all three runs, as everywhere (main §3.2).

Cost scaling, measured. The persistence step — the refresh’s dominant cost — measured on the torus cloud: 174 ms at $M=2048$ (reproducing §3.2’s recorded refresh figure), 388 ms at 4096, 740 ms at 8192, 1.92 s at 20,000

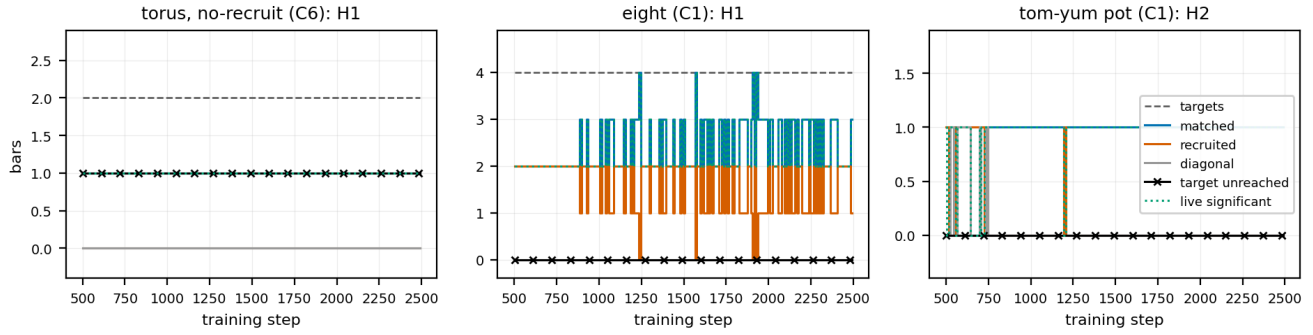


Figure S11. Per-refresh plan composition from the recorded refresh logs (seed 0; ~ 200 refreshes each; no new runs). Left: torus without recruitment (C6) — one loop matched, one target unreachable at every refresh: the chronic gap behind the C6 collapse (main §5.1). Middle: eight (C1) — recruitment claims the two sub-threshold loops early and hands them to the matched set as training makes them significant. Right: the tom-yum pot (C1) — a stable one-bar matched plan after the first ~ 250 active steps. Dashed gray = target count; in the right panel it coincides with the matched line.

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